

ALBERTUS BERNAT SENTOSA

MECHANICAL ENGINEERING GRADUATE | ROBOTICS, AUTOMATION & ADDITIVE MANUFACTURING

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PROFILE

Mechanical Engineering graduate from Atma Jaya Catholic University (GPA 3.88/4.00, *cum laude*; Best Graduate Ring Award). Experienced in independent software development for five-axis 3D printing, robotic-arm prototyping, and experimental research on hydrogen fuel cells. Interested in robotics, manipulation and control, human–robot interaction, and intelligent automation.

EDUCATION

Atma Jaya Catholic University of Indonesia

Bachelor of Mechanical Engineering (S.T.) - November 2021 - July 2026

- GPA: 3.88 / 4.00 - *cum laude*
- Best Graduate Ring Award – top graduate in the Mechanical Engineering program
- Thesis defense grade: A
- Relevant coursework: Robotics, Automatic Control, Mechatronics, Numerical Analysis, Linear Algebra, Finite Element Methods, and PLC.

RESEARCH & PUBLICATION

Undergraduate Thesis – DEVELOPMENT OF 3D MODEL SLICER FOR A 5-AXIS 3D PRINTER | 2025–2026

- Independently developed the complete slicer software, from algorithm research and segment-based non-planar slicing to G-code transformation and GUI implementation.
- Integrated PrusaSlicer and developed matrix-based coordinate transformations for five-axis printing.
- Evaluated physical prints, achieving average dimensional deviation of 0.29 mm (0.95%) and angular deviation of 0.5° in reported measurements.

Published Research – Hydrogen Fuel Cell: Experimental Study of the Generated Voltage | 2025

- Arbiyani, F., Sentosa, A. B., & Limmuel, E. B. JST (Jurnal Sains dan Teknologi), 14(3), 630–642.
- Conducted background research on hydrogen fuel-cell principles and electrochemical reactions.
- Prepared and typed the initial research manuscript; final revisions were made by the supervising professors.
- Independently sourced and purchased materials for the experimental research.
- DOI: https://doi.org/10.23887/jst-undiksha.v14i3.102174

INDUSTRIAL EXPERIENCE

Robotics and Automation Intern – PT Cyber Teknologi Nusantara

July – September 2025

- Worked on an EV charging automation prototype using a Kinova Gen3 6-DoF robotic arm and Kinova-based control software.
- Developed an application for the prototype, designed and 3D-printed the charging station, vehicle charging-port model, and charging gun for simulation.
- Supervisor evaluation: Baik Sekali (Very Good) for fieldwork and internship report.

AWARDS & OTHER ACHIEVEMENTS

- Best Graduate Ring Award, Atma Jaya Catholic University, 2026.
- 1st Place, Best Prototype, and Best Poster, UTM TechVenture Challenge, Universiti Teknologi Malaysia, 2024.
- 2nd Place, Product Innovation Competition, Atma Jaya Business and Innovation Expo, 2024.
- Invited Speaker, 3D Printing Career Day, SMA Kristen Penabur Harapan Indah Bekasi, 2025.
- Best Rank Admission Scholarship, Atma Jaya Catholic University, 2022 – 100% tuition-fee reduction.

TECHNICAL SKILLS

- Programming: Python, C++, Scilab, GNU Octave.
- Robotics & Automation: Kinova Gen3, robotic kinematics (forward/inverse kinematics, Denavit–Hartenberg), Siemens LOGO! Soft Comfort (PLC ladder programming), Arduino.
- Design & Manufacturing: SolidWorks (~3 years), Autodesk Fusion, Tinkercad, PrusaSlicer, FDM and five-axis 3D printing.
- Simulation: ANSYS (introductory experience).

LANGUAGE

Indonesian (Native) | English (CEFR C1) | Japanese (JLPT N5 passed)